

Water Quality Monitoring System: A Smart City Application With IoT Innovation

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Abstract— *In the current era of globalisation, water pollution has gradually worsened, and it has started to reach alarming heights. Water pollution has a severe impact on human health and the environment. The rationale of this research is to propose an affordable water quality monitoring system to save and conserve the quality of consumed water. Water conservation is among the most critical segments of a future smart city project that many countries strive to achieve. This research investigates Malaysia's current water pollution situation and reviews the existing policies that the government has already implemented to curb this issue. Furthermore, it provides an in-depth analysis of similar systems that other fellow researchers have presented. In this paper, the authors have proposed an enhanced water quality monitoring system by implementing an IoT based technological infrastructure suitable for a smart city project. The proposed approach provides a low-cost and power-saving alternative to commercially available systems.*

Keywords — *Water Quality Monitoring, Water Pollution, Water Conservation, Water Quality Monitoring using Smart Devices, Real-time Water Quality Monitoring.*

I. INTRODUCTION

Chemical wastes and oil spills take up significant roles in threatening the country's waterways [1]. With the industrial sector being one of the main culprits in polluting the country's waters, many other industries such as the agricultural sector and many more are affected by the pollution. Furthermore, animals residing in rivers or waterways are constantly plagued by polluted waters, deteriorating their health, and even causing death, which further pollutes the waters. These polluted waters may end up being supplied in domestic houses, threatening the lives and health of the common public. Moreover, lack of civic sense is also one of the causes of water pollution as littering and dumping trash into the water has become a norm. This current scenario is made even more complicated by the lack of modern ways of monitoring, observing, and making rapid decisions in response to changes in water quality. As a result, various countries worldwide have started taking actions to instil awareness towards water quality [2-3].

Current water monitoring systems face multiple challenges. Successfully deployed systems must constantly undergo repairs and maintenance, which can be costly and eat up big budgets. Sensors also need to be very frequently cleaned to ensure that the readings are accurate and not erroneous. In addition, due to the equipment being constantly exposed to water, rust, corrosion, and other maintenance requirements are common issues to be tackled. In addition,

current water monitoring systems are expensive to build or even maintain. The use of old instruments turns out to be highly costly and requires substantial maintenance from time to time. These costs can build up to a significant burden on the users. Finally, users have not been able to view the live feed of the observed location directly.

In this technological era, water quality management systems should interpret readings obtained from the water and analyse these for decision making. At the same time, the system should alert its users when specific readings reach alarming levels, which the user can also preset. An efficient way to monitor water quality is to create an IoT-based water quality monitoring system to curb current water pollution issues. Using IoT, automation can be implemented within the system, decreasing the system's dependency on human resources. Sensors that act as the system's end node will automatically take and send readings to be displayed and shown to the end-user. Actions can be taken immediately in the event of sudden drastic changes in the water quality. Readings from sensors can also be programmed to be sent to the user's mobile devices, and steps can be taken as soon as alarming readings start displaying. Real-time monitoring should be the new approach taken with the help of IoT based water quality monitoring to ensure that the water flowing into common people houses are safe [4].

This research aims to provide a solution to this problem by implementing IoT based technological infrastructure. With the utilisation of modern instruments, efficiency can be increased even with a low budget and maintenance costs. Therefore, the proposed system emphasises using low-cost and easily accessible parts to provide an efficient solution to this problem. With the help of this system, users may obtain a live video feed of the location with just a click of a button.

II. LITERATURE REVIEW

There have been many cases of water pollution happening throughout the years locally, with the percentage of polluted rivers in Malaysia staying at a steadily high rate, as shown in Fig. 1. Referring to the graph, water pollution of the rivers has been a problem since 2008, with percentages of slightly polluted rivers going up for the past few years. Polluted rivers have also increased in percentage from 2015 to 2017, with unchanged numbers of rivers. This shows that the water pollution has been getting more severe throughout the years, even when the number of rivers has reduced from 579 in 2008 to 477 in 2017 [5], a total of 51 rivers were contaminated. According to the water quality index classification shown

below, 24 out of 51 rivers require intensive treatment, 26 are used for irrigation, and one is under dire pollution conditions. A news article from News Straits Time published in 2019 has written that 207 out of 477 rivers were slightly polluted, with another 51 of them being polluted.

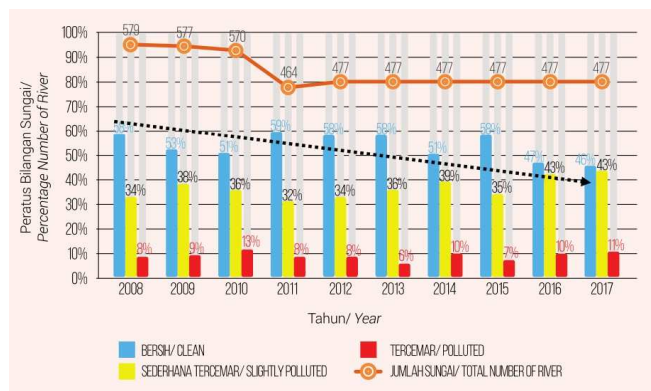


Fig. 1. River Pollution in Malaysia [5]

Water pollution of local waters in Malaysia has stemmed from 5 main sources: manufacturing industries, agricultural-based industries, sewage treatment plants, piggery, and wet markets [6]. Based on existing research, 40% of overall polluting industries come from food and beverage factories, with rubber-producing premises and chemical producers taking up 25% of the total polluting industries [7]. With industrial areas mainly distributed in Selangor, Johor, Penang, Kedah and Perak, these states are bound to have high pollution rates amongst their waters. It is estimated that around 80% of total industrial wastes in Malaysia originate from the above industries mentioned.

One of the main factors that directly cause pollution of water is illegal chemical waste dumping. In the news article reported by New Straits Time in 2019, a water pollution tragedy happened in Sungai Kim of Johor where toxic waste contaminated the river, resulting in airborne contaminants that plagued nearby residents. Up to 6000 people had breathing difficulties after fumes from chemical wastes were dumped illegally into the river [8], forcing nearby schools to shut down to keep the students safe when the incident happened in March of 2019. All 111 schools within the Paris Gudang district were closed until further notice immediately after the incident occurred [9], showing how much of an emergency and disaster this chemical waste dumping was towards the citizens living in Pasir Gudang.

Another factor that causes water pollution in Malaysia would be the negligence and corruption of involved staff regarding the cleanliness of the water they are responsible for. According to an article published by The Straits Times on the 4th and 5th of September 2020 [10], four plant managers were arrested after the water supply cut in Selangor and Kuala Lumpur due to water pollution. Frustrations were vented through social media over unwarned and unscheduled water supply cuts in certain parts of Selangor and Kuala Lumpur [1].

Four water treatment plants were forced to cease operations due to odour pollution, which raised frustrations among people who lived in a very concentrated housing area of Selangor that was required to sort of "compete" for a necessity. In March, the same treatment plant was also fined RM60,000 for a similar offence, causing public anger to boil

over the authorities governing the plants. The incident in March saw the plant releasing heavy machinery oil into local drains that ultimately flow into the Sungai Gong River, heavily polluting the river as a result [11].

The discharged contaminated water coming from industrial areas, sewages and rainwater can be used by people from the agricultural sector, which brings to the first big impact of water pollution. It is estimated that more than 20 million hectares are irrigated with contaminated or partially contaminated water worldwide. It would take up to months to discover the toxins from polluted water ingested by animals or humans. Directly drinking the dirty water will result in typhoid fever or cholera; humans would encounter issues with their hearts or kidneys after being in contact with products that were grown with contaminated water [12].

A survey was conducted in Gombak during 2014 to find out the perceived risk of water contaminations in households and examine the factors that influenced the surveyed perception of the water contamination risk. Respondents were requested to rate six specific diseases caused by water pollution within the Gombak River area. Results indicated that 45.2% of surveyed personnel had chosen diarrhoea as the most severe disease recorded. Water pollution can also cause water to stagnate, which indirectly contributes to the perfect place for the Aedes mosquitoes to reproduce. They also asked respondents for the source of their drinking water, which after collecting answers, it was found that 40% of respondents living in that area were buying drinking water. 30% were said to have obtained drinking water from city water. 16% of respondents said they had individual wells and 6% from community wells to obtain their drinking water. 5% of respondents did not know where their drinking water came from, and the last 3% of respondents got their drinking water from the river [12].

A study was conducted by [13] in two areas: Mukim Part Lubok and Parit Raja in Batu Pahat, Malaysia, to determine the aluminium concentration in drinking water and whether it directly affected the health of an individual. A total number of 100 respondents participated in the study according to the exclusive criteria set. Studies showed incredible potential for adverse health effects that would occur if there were aluminium intakes when drinking water. Therefore, in conclusion, it was necessary to reduce aluminium levels in the drinking water of both areas [13].

Malaysia has been relatively proactive in creating solutions and implementing policies to observe and handle water pollution in Malaysia. Legislations were formed to ensure that enough pressure is placed on private organisations or individuals to emphasise water quality management to tackle water pollution in local waters. The National Policy on Environment states that Malaysia will utilise environmentally friendly and sustainable development approaches to move forward as a country [14]. The policy outlines the strategies and measures that must be implemented to move towards an efficient way of managing water resources, pollution control and the prevention of environmental deterioration. River water quality management requires a holistic approach [14]. Legislations formed under the legislative approach to managing water quality are as follows:

- Environmental Quality (Prescribed Premises)
- (Crude Palm Oil) Regulations: 1977

- Environmental Quality (Sewage and Industrial Effluents) Regulations 1979:
- Environmental Quality (Prescribed Premises)
- (Raw Natural Rubber) Regulations 1978

The formation of these legislations will help control pollution and act as a reference when a particular individual or organisation conducts water pollution. Other than the legislation's formation to protect and preserve the water quality of local waters, the government has made conducive changes to their organisational arrangement related to water quality management. For example, the Ministry of Natural Resources & Environment was formed in 2004 by combining ministers from four other departments to ensure the integrity in the management of water resources. Furthermore, certain ministers were tasked to supervise certain matters on water resources and based on the policy statement. The objectives of national policy on the environment are:

- To create a clean, safe, healthy, and productive environment for both current and coming generations
- To preserve the country's unique natural heritage with active participation by all sectors of society
- To create a sustainable lifestyle & pattern

The government has also established the National Water Services Industry Commission (SPAN) under the provision of the Water Services Commission Act. It has employed the WSIA as regulatory tools for regulating the water services industry, such as water treatment plants in monitoring, licensing, and supervision. The commission would serve as the central regulatory body (middleman) as State Governments would have to corporatise their state water authorities. It will regulate all entities in the sewerage services and water supply industry regardless of public or private water supply. Entities regulated by SPAN as shown in Fig. 2. [15].

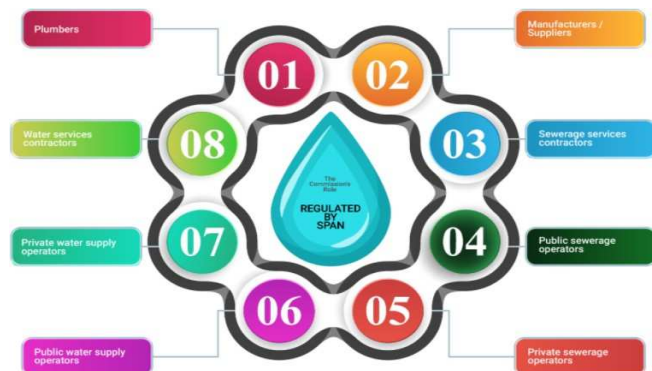


Fig. 2. Entities Regulated by SPAN [15]

III. SIMILAR SYSTEMS REVIEW

In this section, the researchers have reviewed similar systems proposed by other researchers on the water quality monitoring system. These systems have been proposed considering various local factors in mind, and each has its own way of implementation and using structural components.

A. Water Quality Monitoring System based on Internet of Things (IoT)

Addressing the seriousness of China's water pollution, a technologically advanced water quality system has been proposed. With the implementation of IoT, previous manual

recording and sampling that were tedious and ineffective are proposed to be replaced with automation. The system utilises the STM32 controller for data processing and transmission to a remote management platform concurrently via WIFI to the personnel responsible for observing water quality. The main functions of the system include real-time water quality monitoring, statistical charts, and abnormal state alarm. Different layers of design for the software have also been explained in detail.

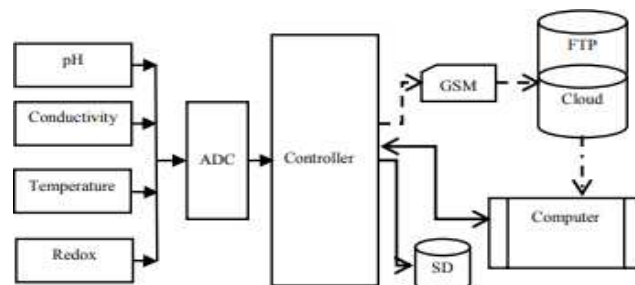


Fig. 3. Structural Design of the Water Quality Monitoring System based on Internet of Things [3]

From the Fig. 3 structural design of the system, the outermost layer will be the sensing layer comprised of temperature sensors, turbidity sensors and PH sensors. All the obtained readings from the sensors on the sensing layer will be sent to the main controller in the control layer, the STM32 microcomputer. The STM32 microcomputer coordinates and controls the reading, transmission, receiving and processing of sensor data from the sensing layer. The processed readings are then sent through the network transport layer by Wi-Fi to a remote cloud management centre. The ESP8266 Wi-Fi module would provide the microcomputer access to a Wi-Fi network, providing a connection between the hardware and software. The user will be able to configure thresholds for the alarm through a mobile application. After sending the data to the cloud management centre through Wi-Fi, the system will monitor all data received and store it in a SQL Server database that will establish two databases: historical database and interactive database. The historical database is responsible for storing past monitored data and readings from the sensor layer; meanwhile, the interactive database only stores one piece of data at a time, which directly comes from the sensor layer through the network layer in real-time. The interactive database is also responsible for interacting with the application (web page) and the cloud server. The application provides users with real-time readings from the sensor layer by reading data from the interactive database [3].

B. Smart Water Quality Monitoring System

Smart Water Quality Monitoring System, proposed by [1], obtains readings in pH, Oxidation and Reduction Potential (ORP), Conductivity and Temperature by using sensors by implementing RS technology. The area of this research is in Fiji, where water pollution is caused primarily by chemical waste and oil spills. Fig. 4 shows the block diagram of the system that comprises PH, conductivity, temperature, and redox sensors. These sensors will send their readings to an analogue to digital converter (ADC) before reaching the microcontroller.

The data in this system is stored in two ways, SD and GSM. The SD would be physical storage that allows the storage of obtained readings, whereas data will be saved in a

cloud server. The system also incorporates a power-saving mode that allows the system to go into a sleep mode after an hour of consecutive readings to extend battery life. Idle modules will also be turned off to conserve battery. Alerts regarding battery life or reports will be sent to the user [1].

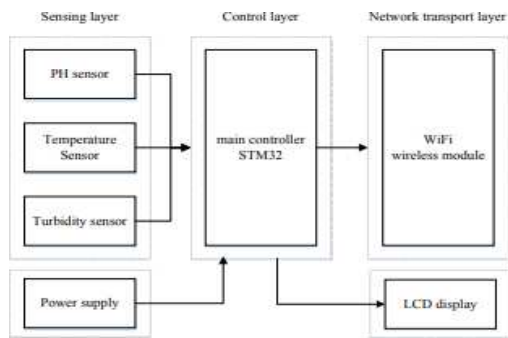


Fig. 4. Structural Design of the Smart Water Quality Monitoring System [1]

C. Reconfigurable Smart Water Quality Monitoring System in IoT Environment

A reconfigurable smart water quality monitoring system in IoT environment was proposed by [16]. The proposed system will be equipped with Field Programmable Gate Array (FPGA) design board, a sensor layer, a Zigbee based wireless communication module and a computer for information display. The FPGA board will act as the system's core component, being programmed in very high-speed integrated circuit hardware description language (VHDL) and C programming language while utilising its respective tools and software. The proposed system will collect five different water parameters, including pH, turbidity, water level, carbon dioxide (CO₂), and water temperature, on a real-time basis as shown in Fig. 5.

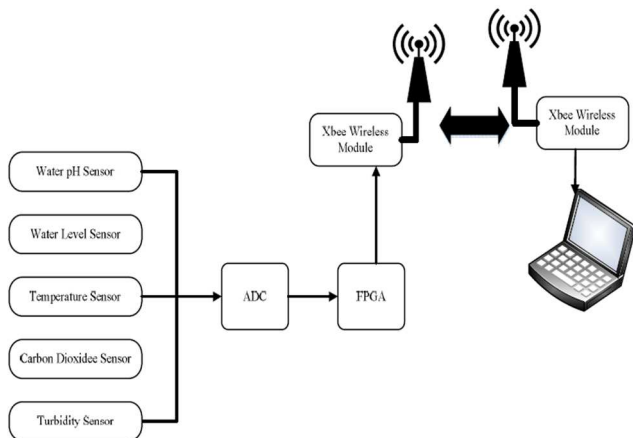


Fig. 5. Structural Design of the Reconfigurable Smart Water Quality Monitoring System in IoT Environment [16]

D. Comparison of Water Quality Monitoring Systems

Table 1 provides a comparison of the discussed three systems in terms their functionalities. The factors considered for comparison include real-time data monitoring, alert when readings reach certain thresholds, configurable alert thresholds, storage methods, display of information, additional features, measured parameters and microcontroller used.

TABLE I. COMPARISON OF THE SIMILAR SYSTEMS

	Water Quality Monitoring System Based on Internet of Things	Smart Water Quality Monitoring System	Reconfigurable Smart Water Quality Monitoring System in IoT Environment
Real-time data monitoring	Yes	Yes	Yes
Alert when readings reach certain thresholds	Yes	No Alert mechanism for readings	Yes
Configurable alert thresholds	Yes	No	Yes
Storage Methods	Cloud	Local SD and cloud	Cloud
Display of information	Mobile application	Computer	Computer
Additional features	None	Power saving mode	None
Measured parameters	pH, Temperature, Turbidity	pH, Conductivity, Temperature, Redox	pH, water level, water temperature, carbon dioxide, turbidity
Microcontroller used	Arduino Atmega	STM32 embedded microcontroller	Waspnote Microcontroller board

IV. PROPOSED SYSTEM ARCHITECTURE

The researchers proposed an IoT-based system that provides a daily report and generates an email alert if any catastrophe occurs. There are two users in the login. One is admin, and the other is a normal user. The admin has full access to control the entire system. He can upload the codes into the Arduino Mega board, which control all the sensors, and the codes will be executed accordingly. The sensors will be configured to obtain readings and then be sent through Node-Red inserted into the InfluxDB database.

As most of the output from the devices are in JSON form, it allows the nodes in Node-Red to capture the correct reading according to the variables assigned to each reading obtained. Node-Red will be responsible for transferring the readings obtained into the InfluxDB database. Readings from the Serial monitor will start to be received when the board is connected; the output is then transferred to a JSON node followed by functions that capture the respective readings to be inserted into the InfluxDB database called "readings". The database created is also named readings. This will also be the only database utilised in the system, and it will consist of all the readings and data captured by the sensors. The preset command "influx -precision rfc3339" is inputted to change the format of the Date and Time that will be automatically recorded upon receiving a reading.

The proposed system allows admin users to add/remove sensors, increase/decrease the thresholds, increase/decrease the sensitivity of the sensors, view readings in InfluxDB, configure nodes, view graphs with readings in Grafana, add/remove observations in Grafana and adjust the positions of the sensors placed in the observed area. On the other hand, the normal user is only able to view sensor readings and

thresholds in Grafana and see the outcome when certain readings exceed the preset threshold as shown in Fig. 6.

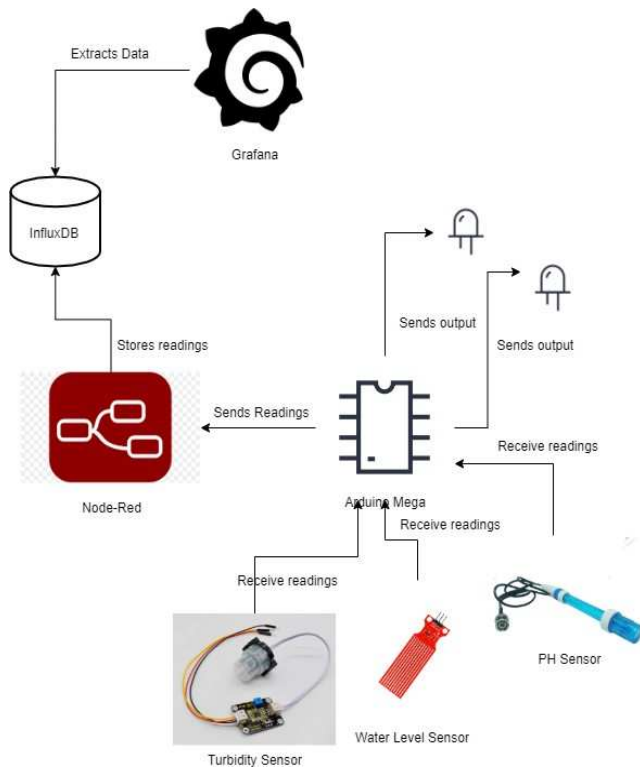


Fig. 6. Architectural Design of the Proposed IoT based Water Quality Monitoring System for Malaysia

Fig. 7 represents the water quality monitoring using Grafana Dashboard. There are three graphs displayed, which are PH_Value, Voltage (turbidity) and Water Level. Readings are displayed in the green line, whereas the red line indicates the threshold for the particular reading. If readings go over the preset threshold consistently for 10 seconds, a brown dotted vertical line is displayed indicating that the alert is pending, followed by a red dotted vertical line after ten more seconds of monitoring, which indicates that an Email has been sent to the inbox of the user.



Fig. 7. Water Quality Monitoring Using Grafana Dashboard

V. CRITICAL ANALYSES AND DISCUSSION

Creating and developing a water quality monitoring system requires few crucial phases to be completed. The phases are change detection, data insertion and extraction, and display of information. The system's user interface should be user-friendly as it will help the end-users familiarise themselves with the system quickly. The interface and its functions should be relatively straightforward and require minimal or no configuration at all. A user-friendly user interface would certainly increase the satisfaction of end-users. Change detection is the ability of the sensors to detect changes within the water. It was a challenging task for the researchers to find a suitable place to set up the water quality monitoring system to ensure no other external objects or phenomena that may affect the change detection of the sensors. Moreover, sensors require configurations to obtain appropriate readings and perform certain functions according to project requirements. Data insertion occurs when the collected data or readings from the sensors are inserted into a real-time database. Extraction of data then occurs when the information is extracted from the database to display live water quality.

Display of information is about displaying the readings along with the calculated water quality index. Readings should be displayed to the user in real-time to allow users to take action or be updated as soon as possible upon receiving alerts. Therefore, a user interface must be designed, which causes extra pressure and burden on the programming part of the entire project. However, programming could be reduced or made obsolete with the help of IoT application development platforms such as Blynk, which will be elaborated in detail after. It could still pose a challenge to the developer because the developer would first need to learn the ropes of utilising the platform before developing anything with the application.

The researchers have discovered certain limitations during the process of developing the water quality monitoring system. First and foremost, the sensors being utilised to obtain readings from the water are not fully waterproof. The sensors might need further waterproofing if placed on the observation site for long periods. Therefore, external measures still need to be taken to apply waterproofing on the sensors, the board, and the computer, which brings about another limitation. Besides, since the sensors and other devices are not completely waterproofed, they are only suitable to be placed in the shallow part of the waterway.

Overall, the proposed system will be much cheaper than the available commercial water quality monitoring devices. The approximate prices of the component are as follows:

- Arduino Mega (Non-official board): RM30
- Cables, wires, resistors, and LEDs: RM15
- PH Sensor with Buffer Solution: RM250
- Turbidity Sensor: RM70
- Water Level Sensor: RM5
- Breadboard: RM3

The total price adds up to around RM375. If the user wishes to use a more refined (industrial scale) water level sensor, that will cost four times more than the price of this setup.

VI. CONCLUSION

This research has been conducted to figure out the issues, key factors, and solutions to curb the water pollution issue. The proposed solution is a cost-effective and efficient alternative to the existing commercial water quality monitoring solutions. It relies on the concept of IoT to ease out the communication between sensors and other devices. Future research must extend this work using fully waterproof sensors to do more reliable system testing. Moreover, a live feed function can be added, allowing users to see a live video feed of the observed area.

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